ID No: ____________________Name: ___________________________________Sec No.: ________

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

II SEMESTER 2025-26

A

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

28-01-2026 (Friday) QUIZ-1 (closed book)

Max time: 15 Min Max Marks: 08

Q1. (a) What is the size of a segment that can be accessed by 8086 processors. Assume memory is byte organized. Answer should be only mentioned in KB? [**1M]**
(b)What is the starting address of the segment if the ending address is AE000? **[1M]**

Calculation	**Final Answer**
	(a)64KB **(b) 9E001H**

Q2. Consider the following 8086 assembly instruction: SUB AX, BX (Instruction subtracts BX from AX and stores the result in AX, along with the updation of flags). Assume that the condition of all the status flags of the flag register is cleared (RESET) before the execution of the instruction.
AX = 7B3FH
BX = 84C1H
After executing the instruction, determine the **status (SET(1) or RESET (0))** of the following flags:
(1) Parity Flag
(2) Carry Flag

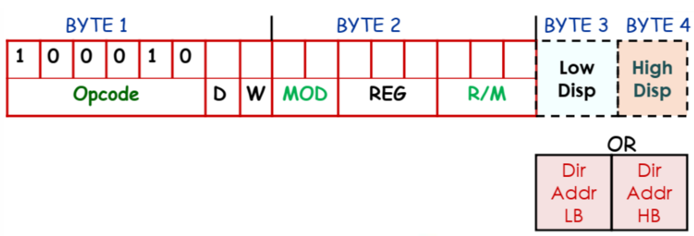
Calculation	**Final Answer**
	PF=1 **CF=1**

Q3. For the 8086 processor, only one following instructions is valid.
MOV BX, [AX]
MOV AL, ES: [SI+10H]
MOV [BP+CX+20H], AL
What is/are the physical address(s) accessed via the valid instruction [2M]
AX=2000H, SP=2302H, BP=0402H, SI=1ABCH, DI=1A56H, DS=1000H, CS=2000H, SS=3000H, ES=4000H

Calculation	**Final Answer(s)**
	41ACCH

Q.4. Find the machine code for the instruction MOV [BX+10H], AX for the 80386 processor operating in 32-bit mode. Write the machine code in hexadecimal only. Show Calculation. [2M]

Calculation	**Final Answer (in hexadecimal)**
	67 66 89 47 10



EAX/AX/AL	000
EBX/BX/BL	011
ECX/CX/CL	001
EDX/DX/DL	010
ESP/SP/AH	100
EBP/BP/CH	101
ESI/SI/DH	110
EDI/DI/BH	111

REG

D = 0 (Direction from Reg)
= 1 (Direction to Reg)

W = 0 (Data - byte)
= 1 (Data - word)

MOD / R/M	00	01	10	11 W = 0	11 W = 1
000	[BX] + [SI]	[BX]+[SI] +d8	[BX]+[SI]+d16	AL	AX
001	[BX] + [DI]	[BX]+[DI]+d8	[BX]+[DI]+d16	CL	CX
010	[BP] + [SI]	[BP]+[SI]+d8	[BP]+[SI]+d16	DL	DX
011	[BP] + [DI]	[BP]+[DI]+d8	[BP]+[DI]+d16	BL	BX
100	[SI]	[SI]+d8	[SI]+d16	AH	SP
101	[DI]	[DI]+d8	[DI]+d16	CH	BP
110	d16	[BP] + d8	[BP] + d16	DH	SI
111	[BX]	[BX]+d8	[BX]+d16	BH	DI

MOD / R/M	00	01	10	11 W = 0	11 W = 1
000	EAX	EAX+d8	EAX+d32	AL	EAX
001	ECX	ECX+d8	ECX+d32	CL	ECX
010	EDX	EDX+d8	EDX+d32	DL	EDX
011	EBX	EBX+d8	EBX+d32	BL	EBX
100	Scaled Index	Scaled Index +d8	Scaled Index +d32	AH	ESP
101	d32	EBP+d8	EBP+d32	CH	EBP
110	ESI	ESI+d8	ESI+d32	DH	ESI
111	EDI	EDI+d8	EDI+d32	BH	EDI

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
II SEMESTER 2025-26
CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing
30-1-2026 (Friday) QUIZ-1 (closed book)

B

Max time: 15 Min **Max Marks: 08**

Q.1. (a) What is the size of primary memory that can be accessed by 80236 processor. Assume memory is byte organized. Answer should be only mentioned in MB? [**1M]**
(b)What is the starting address of segment if the ending address is 3 4E23h? **[1M]**

Calculation	**Final Answer**
	(a)16MB **(B)**24E24H

Q2. Consider the following 8086 assembly instruction: SUB AX, BX (Instruction subtracts BX from AX and stores the result in AX along with the updation of flags). Assume that condition of all the status flags of the flag register are cleared (RESET) before the execution of the instruction.
AX = 7B3FH
BX = 84C1H
After executing the instruction, determine the **status (SET(1) or RESET (0))** of the following flags:
(1) Auxiliary Carry Flag
(2) Overflow Flag [1M+1M]

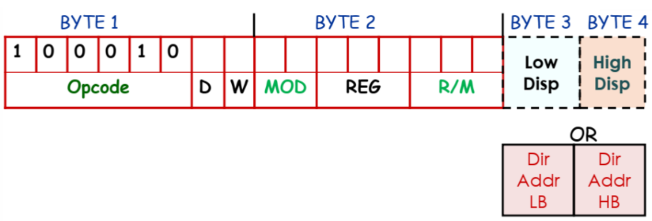
Calculation	**Final Answer**
	AF=0 **OF=1**

Q3.For 8086 processor, only one of the instruction is valid.
MOV DS, [AX]
MOV AL, [SP+10H]
MOV [BP+20H], AL
What is/are the physical address(s) accessed via the valid instruction [2M]
AX=2000H, SP=2302H, BP=0402H, SI=1ABCH, DI=1A56H, DS=1000H, CS=2000H,SS=3000H, ES=4000H

Calculation	**Final Answer**
	30422H

Q.4. Find the machine code for the instruction MOV [ESI+10H], EAX for the 80386 processor operating in 16-bit mode. Write the machine code in hexadecimal only. Show Calculation. [2M]

Calculation	**Final Answer (in hexadecimal)**
	67 66 89 46 10



D = 0 (Direction from Reg)
= 1 (Direction to Reg)

W = 0 (Data - byte)
= 1 (Data - word)

EAX/AX/AL	000
EBX/BX/BL	011
ECX/CX/CL	001
EDX/DX/DL	010
ESP/SP/AH	100
EBP/BP/CH	101
ESI/SI/DH	110
EDI/DI/BH	111

REG

MOD R/M	00	01	10	11 W = 0	11 W = 1
000	[BX] + [SI]	[BX]+[SI] +d8	[BX]+[SI]+d16	AL	AX
001	[BX] + [DI]	[BX]+[DI]+d8	[BX]+[DI]+d16	CL	CX
010	[BP] + [SI]	[BP]+[SI]+d8	[BP]+[SI]+d16	DL	DX
011	[BP] + [DI]	[BP]+[DI]+d8	[BP]+[DI]+d16	BL	BX
100	[SI]	[SI]+d8	[SI]+d16	AH	SP
101	[DI]	[DI]+d8	[DI]+d16	CH	BP
110	d16	[BP] + d8	[BP] + d16	DH	SI
111	[BX]	[BX]+d8	[BX]+d16	BH	DI

MOD R/M	00	01	10	11 W = 0	11 W = 1
000	EAX	EAX+d8	EAX+d32	AL	EAX
001	ECX	ECX+d8	ECX+d32	CL	ECX
010	EDX	EDX+d8	EDX+d32	DL	EDX
011	EBX	EBX+d8	EBX+d32	BL	EBX
100	Scaled Index	Scaled Index +d8	Scaled Index +d32	AH	ESP
101	d32	EBP+d8	EBP+d32	CH	EBP
110	ESI	ESI+d8	ESI+d32	DH	ESI
111	EDI	EDI+d8	EDI+d32	BH	EDI